

MAT 549 HW05

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Problem 1 (15-1). Suppose M is a smooth manifold that is the union of two orientable open submanifolds with connected intersection. Show that M is orientable. Use this to give another proof that S^n is orientable.

Solution. Let U and V be the two open submanifolds and let ω and η respectively be orientation forms on U and V . Since $U \cap V$ is connected, we can assume that the restrictions of ω and η to $U \cap V$ induce the same orientation on $U \cap V$ by negating η if necessary. Let $\{\varphi_U, \varphi_V\}$ be a partition of unity subordinate to the open cover $\{U, V\}$ of M (so φ_U is zero outside U and φ_V is zero outside V); then I claim that $\varphi_U \cdot \omega + \varphi_V \cdot \eta$ is an orientation form on M . By definition this form is nonzero on $U - U \cap V$ and $V - U \cap V$, and for all $p \in U \cap V$ and $e_1, \dots, e_n \in T_p M$, $\omega_p(e_1, \dots, e_n)$ has the same sign as $\eta_p(e_1, \dots, e_n)$, so $\varphi_U(p)\omega_p(e_1, \dots, e_n) + \varphi_V(p)\eta_p(e_1, \dots, e_n)$ has the same sign as the values of ω and η , and in particular is nonzero. $\square$

Problem 2 (15-3). Suppose $n \geq 1$, and let $\alpha: S^n \rightarrow S^n$ be the antipodal map $\alpha(x) = -x$. Show that α is orientation preserving if and only if n is odd.

Solution. The vector $p \in S^n \subseteq \mathbb{R}^{n+1}$ is a unit normal to $T_p S^n$, so the normal vector field $X_p = p$ is a nonvanishing section of the rank-1 normal bundle of S^n . The existence of this section tells us that S^n has trivial normal bundle in $\mathbb{R}^{n+1}$.

Let ω be the volume form $dx_1 \wedge \dots \wedge dx_n$ on $\mathbb{R}^{n+1}$; by the solution to Problem 15-8(a) below, $\iota_X \omega$ is a volume form on S^n . Then α is orientation-preserving if and only if $\alpha^*(\iota_X \omega)$ induces the same orientation on S^n as $\iota_X \omega$ (in general this holds for any diffeomorphism α , since we want the differential at each point to induce an orientation preserving linear map on the top exterior product of each tangent space).

Let $r_i: \mathbb{R}^{n+1} \rightarrow \mathbb{R}^{n+1}$ be the reflection map $r(x_1, \dots, x_i, \dots, x_n) = r(x_1, \dots, -x_i, \dots, x_n)$, so α is the composition of all the $r_i|_{S^n}$; of course, each r_i is a diffeomorphism and restricts to a diffeomorphism on S^n (we will identify r_i and $r_i|_{S^n}$). Each r_i reverses the orientation because for all $p \in S^n$ and $e_1, \dots, e_n \in T_p S^n$ a basis,

$$(r_i^*(\iota_X \omega))_p(e_1, \dots, e_n) = \omega_p((dr_i)_p X_p, (dr_i)_p e_1, \dots, (dr_i)_p e_n) = -\omega_p(X_p, e_1, \dots, e_n) = -(\iota_X \omega)_p(e_1, \dots, e_n).$$

The last inequality is true because $\det(dr_i)_p = -1$ and for a finite-dimensional vector space V and linear map $T: V \rightarrow V$, the induced map on the top exterior product space $\Lambda^{\dim V} V$ is multiplication by the determinant. Finally, we find that

$$\alpha^*(\iota_X \omega) = r_1^* \dots r_{n+1}^*(\iota_X \omega) = (-1)^{n+1} \iota_X \omega,$$

so α is orientation-preserving if and only if $(-1)^{n+1}$ is positive, or equivalently n is odd. $\square$

Problem 3 (15-5). Let M be a smooth manifold with or without boundary. Show that the total spaces of TM and T^*M are orientable.

Solution. Let $p \in M$ be an arbitrary point, U a neighborhood of p , and $(x_1, \dots, x_n)$ a local coordinate system in a neighborhood of p . These induce local coordinates $(x_1, \dots, x_n, v_1, \dots, v_n)$ on $\pi^{-1}(U)$. Define a local frame on $\pi^{-1}(U)$ by $(X_p^1, \dots, X_p^{2n}) = (\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_n}, \frac{\partial}{\partial v_1}, \dots, \frac{\partial}{\partial v_n})$. If $\varphi: U \cap V \rightarrow U \cap V$ is the coordinate change map between two coordinate charts U and V , then the corresponding coordinate change map between the charts on U and V has differential $d\varphi_p \oplus d\varphi_p$ and Jacobian $\det(d\varphi_p)^2$, so the local frames on all charts defined in this way are consistently oriented, implying TM is orientable.

Every smooth manifold admits a Riemannian metric, which then provides a natural isomorphism between the tangent and cotangent bundles (by the results of Chapter 13), so the cotangent bundle is orientable by virtue of being diffeomorphic to an orientable manifold. $\square$

Problem 4 (15-8). Suppose M^m is an orientable Riemannian manifold and $S^s \subseteq M$ is an embedded or immersed submanifold. Prove that

- (a) If S has trivial normal bundle, then S is orientable.
- (b) If S is an orientable hypersurface, then it has trivial normal bundle.

Solution. (a) Let ω be a volume form on M and $i: S \rightarrow M$ the immersion/embedding that defines S as a submanifold of M . Since the normal bundle N is trivial, it admits $r = \text{rank } N$ linearly independent (global) sections $X_1, \dots, X_r$. Then, consider the s -form

$$\eta = \omega(X_1, \dots, X_r, -) = \iota_{X_1}(\cdots \iota_{X_{r-1}}(\iota_{X_r}\omega))$$

(where ι_V is interior multiplication by the vector field V).

I claim that $i^*\eta$ is a volume form on S . To see this, we note that for all $p \in S$, if $e_1, \dots, e_s \in T_p S$ is a positively oriented basis of $T_p S$, then $di_p e_1, \dots, di_p e_s$ forms a basis of $T_{i(p)} M$ with the $(X_i)_p$ because i is an immersion, so

$$(i^*\eta)_p(e_1, \dots, e_s) = \eta_{i(p)}(di_p e_1, \dots, di_p e_s) = \omega_{i(p)}((X_1)_p, \dots, (X_r)_p, di_p e_1, \dots, di_p e_s) \neq 0.$$

That is, $i^*\eta$ is nonzero at each point of S , so it defines an orientation on S .

(b) For each $p \in S$, let $e_1, \dots, e_s \in T_p S$ be a positively oriented basis of S (with respect to the orientation on S). Let X_p be the unique unit normal vector at p such that $(X_p, e_1, \dots, e_s)$ is oriented positively according to the orientation on M . This is a smooth nonvanishing section of the normal bundle; smoothness follows from taking a positively oriented orthonormal frame around p which extends a positively oriented orthonormal frame for S around p , since this necessarily forces the one vector not in $T_p S$ to equal X_p , and local frames consist of smooth vector fields. The existence of a nonvanishing global smooth section of the normal bundle (which has rank 1) implies that it is trivial. $\square$